

One Hot Encoding

Intuition, then advantages and disadvantages.

Next Job: Convert Text into Vectors

DONE

Text preprocessing: tokenization, stopwords, stemming, lemmatization



NOW

Text → vectors **1. One hot encoding** 2. Bag of words · 3. TF-IDF

Find the Vocabulary (Unique Words)

DATASET

	Text	O/P
D1	the movie is great	1
D2	the movie is boring	0
D3	story is fun	1

VOCABULARY · V = 7

1

the

2

movie

3

is

4

great

5

boring

6

story

7

fun

D1

the, movie, is, great

4 new

D2

boring

1 new, the/movie/is repeat

D3

story, fun

2 new, is repeats

4 + 1 + 2 = 7 unique words → V = 7

Each Word Becomes a Vector of Size V

DATASET

	Text	O/P
D1	the movie is great	1
D2	the movie is boring	0
D3	story is fun	1

VOCABULARY · V = 7

1

2

3

4

5

6

7

the

movie

is

great

boring

story

fun

the	1	0	0	0	0	0	0
movie	0	1	0	0	0	0	0
is	0	0	1	0	0	0	0
great	0	0	0	1	0	0	0
boring	0	0	0	0	1	0	0
story	0	0	0	0	0	1	0
fun	0	0	0	0	0	0	1

A 1 at the word's position, 0 everywhere else.

D1: "the movie is great"

DATASET

	Text	O/P
D1	the movie is great	1
D2	the movie is boring	0
D3	story is fun	1

VOCABULARY · V = 7

1	2	3	4	5	6	7
the	movie	is	great	boring	story	fun

	the	movie	is	great	boring	story	fun
the	1	0	0	0	0	0	0
movie	0	1	0	0	0	0	0
is	0	0	1	0	0	0	0
great	0	0	0	1	0	0	0

```
D1 = [[1, 0, 0, 0, 0, 0, 0],
[0, 1, 0, 0, 0, 0, 0],
[0, 0, 1, 0, 0, 0, 0],
[0, 0, 0, 1, 0, 0, 0]]
```

shape 4 × 7 4 words × 7 vocabulary size

D2: "the movie is boring"

DATASET

	Text	O/P
D1	the movie is great	1
D2	the movie is boring	0
D3	story is fun	1

VOCABULARY · V = 7

1	2	3	4	5	6	7
the	movie	is	great	boring	story	fun

	the	movie	is	great	boring	story	fun
the	1	0	0	0	0	0	0
movie	0	1	0	0	0	0	0
is	0	0	1	0	0	0	0
boring	0	0	0	0	1	0	0

```
D2 = [[1, 0, 0, 0, 0, 0, 0],
[0, 1, 0, 0, 0, 0, 0],
[0, 0, 1, 0, 0, 0, 0],
[0, 0, 0, 0, 1, 0, 0]]
```

shape 4 × 7 4 words × 7 vocabulary size

D3: "story is fun"

DATASET

	Text	O/P
D1	the movie is great	1
D2	the movie is boring	0
D3	story is fun	1

VOCABULARY · V = 7

1	2	3	4	5	6	7
the	movie	is	great	boring	story	fun

	the	movie	is	great	boring	story	fun
story	0	0	0	0	0	1	0
is	0	0	1	0	0	0	0
fun	0	0	0	0	0	0	1

```
D3 = [[0, 0, 0, 0, 0, 1, 0],
      [0, 0, 1, 0, 0, 0, 0],
      [0, 0, 0, 0, 0, 0, 1]]
```

shape 3 × 7

3 words × 7 vocabulary size

D1, D2 and D3 as Python Lists

D1

"the movie is great"

```
D1 = [[1, 0, 0, 0, 0, 0, 0],  
      [0, 1, 0, 0, 0, 0, 0],  
      [0, 0, 1, 0, 0, 0, 0],  
      [0, 0, 0, 1, 0, 0, 0]]
```

4 × 7

D2

"the movie is boring"

```
D2 = [[1, 0, 0, 0, 0, 0, 0],  
      [0, 1, 0, 0, 0, 0, 0],  
      [0, 0, 1, 0, 0, 0, 0],  
      [0, 0, 0, 0, 1, 0, 0]]
```

4 × 7

D3

"story is fun"

```
D3 = [[0, 0, 0, 0, 0, 1, 0],  
      [0, 0, 1, 0, 0, 0, 0],  
      [0, 0, 0, 0, 0, 0, 1]]
```

3 × 7

PART 2

Advantages and Disadvantages of One Hot Encoding

ADVANTAGE

Easy to Implement with Python

SCIKIT-LEARN

OneHotEncoder

PANDAS

pd.get_dummies()

Simple to write, but rarely used in NLP because of the disadvantages that follow.

Sparse Matrix → Overfitting

	the	movie	is	great	boring	story	fun
the	1	0	0	0	0	0	0
movie	0	1	0	0	0	0	0
is	0	0	1	0	0	0	0
great	0	0	0	1	0	0	0

24 of 28 values are 0. Mostly zeros with a few ones is a **sparse matrix**.

Real vocabulary: **50,000** words. Each word is 49,999 zeros and one 1.

Overfitting: great accuracy on training data, poor on new data.

shape 4 × 7 4 words × 7 vocabulary size

ML Algorithms Need a Fixed Size Input

DATASET

	Text	O/P
D1	the movie is great	1
D2	the movie is boring	0
D3	story is fun	1

VOCABULARY · V = 7

1

the

2

movie

3

is

4

great

5

boring

6

story

7

fun

D1

the movie is great

4 × 7

D2

the movie is boring

4 × 7

D3

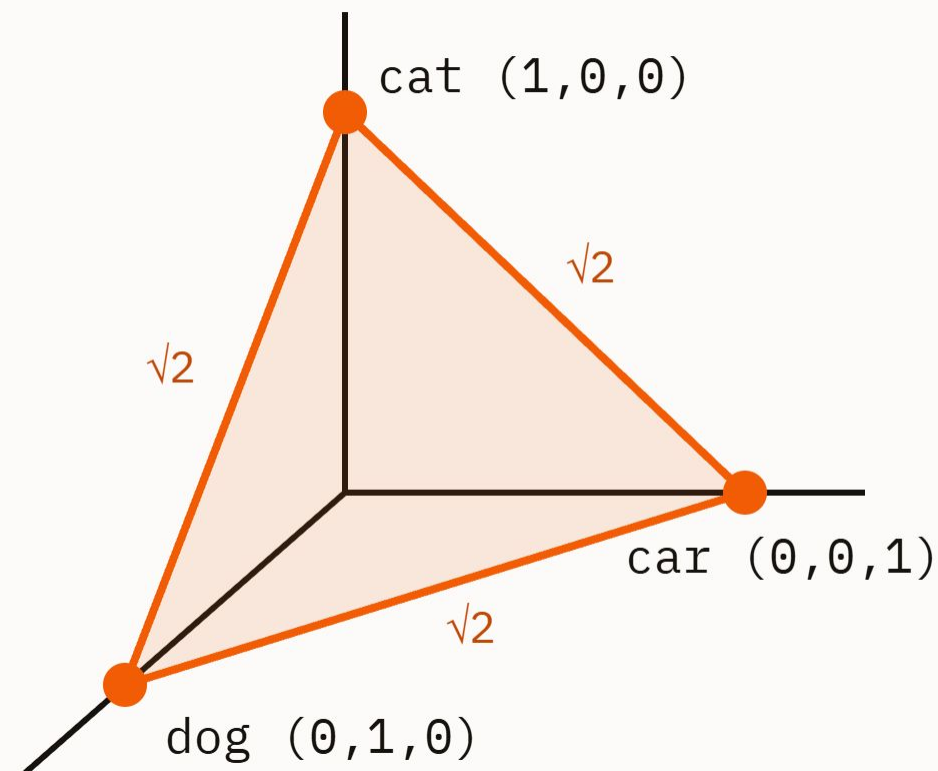
story is fun

3 × 7

Shape depends on sentence length, so the rows do not match.

No Semantic Meaning Is Captured

	cat	dog	car
cat	1	0	0
dog	0	1	0
car	0	0	1



Every pair is the **same distance** apart.

Cat and dog are both animals, but look as different as cat and car.

Out of Vocabulary (OOV)

DATASET

	Text	O/P
D1	the movie is great	1
D2	the movie is boring	0
D3	story is fun	1
Test	the plot is boring	?

VOCABULARY · V = 7

1

the

2

movie

3

is

4

great

5

boring

6

story

7

fun

TEST: "THE PLOT IS BORING"

the1000000

plotNot in vocabulary. No vector possible.

is0010000

boring0000100

SUMMARY

What to Remember

ADVANTAGE

Easy to implement: `OneHotEncoder`,
`pd.get_dummies()`

DISADVANTAGES

- 01 Sparse matrix → overfitting
- 02 No fixed size input for ML
- 03 No semantic meaning captured
- 04 Out of vocabulary (OOV)

Next: Bag of Words, which fixes some of these problems.